

Arnav Rastogi

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Research Interests

Interpretability, Recursive Models, Novel Model Architectures, BitNets.

Education

Indian Institute of Technology, Jodhpur, BS in Applied AI and Data Science

Jan 2025 – Jan 2029

- CGPA: 8.63/10
- Coursework: Deep Learning, Linear Algebra, Statistical Inference.

MITx, MicroMasters in Statistics and Data Science

Sept 2024 – present

- Coursework: Probabilistic Modeling, Machine Learning, Time Series Analysis.

Projects

Learning & Unlearning Noise in State-Space Models (code, write-up)

2026

- Extended a QUEST Lab (IISc) study on noise learning/unlearning in transformer SLMs to **Mamba-1.4B**, running the full three-phase protocol (clean finetune → noisy finetune → clean retrain) on SQuAD across 4 noise types.
- Found Mamba absorbs token-level noise far more than transformers (**71% accuracy drop** under word-flip vs. 27–38% for Olmo/Qwen/Gemma/Phi-2), while counterfactual noise *raised* its accuracy, a behavior no transformer showed.
- Identified confounds limiting the comparison (base vs. instruction-tuned models, fuzzy-match vs. LLM-judge scoring, extra epochs inflating Phase 3 recovery to 124–133%) and documented them alongside results.

ARC-Engine

2023

- Developed a high-performance OpenGL rendering engine in **C++** with a custom **Entity Component System (ECS)** for cache-friendly memory layout.
- Built a real-time visualization pipeline with custom compute kernels for low-level, compute-heavy workloads.

rCiv (Knowledge Commons)

2025 – present

- Built a **FastAPI** service where LLM agents submit source-referenced claims that resolve into a temporal **knowledge graph** (Graphiti + Neo4j), with LLM-based deduplication and conflict detection over hybrid semantic/BM25/graph retrieval.

LLM Lab

Jan 2025 – present

- Implemented a character-level GPT from scratch in **PyTorch** (multi-head attention, residual stream, LayerNorm) as a testbed for architecture experiments.

Technical Writing

Multi-Hop Reasoning in Transformers (link): Jan 2026

ReLU vs GELU (link): Jan 2026

Experience

AI Engineer Intern, Hapticware Intelligence – Pune, India

Mar 2026 – June 2026

- Built an agent orchestration framework with **LangGraph** coordinating 30+ agents across decomposable tasks.
- Engineered **RAG** over a 100+ table production database, cutting chatbot query latency by 30-40% and auto-generating result graphs.

Research Intern, Abhay Travels – India

Feb 2025 – Mar 2025

- Built a **CatBoost** demand-forecasting pipeline with 15+ engineered seasonal features that beat ARIMA baselines on RMSE over 5 years of booking data.

Skills

ML & Research: PyTorch, Transformers, State-Space Models (Mamba), LLM finetuning & evaluation, CatBoost

Mathematics: Linear Algebra, Probability & Statistics, Optimization

Engineering: Python, C++, SQL, OpenGL, FastAPI, Git

Open Source

OmniRoute (PRs)

2026

- 3 merged PRs to a 74K-star open-source LLM gateway, including Zed-hosted provider auth for model discovery, provider wire-format fixes, and repairing two live-server E2E CI suites.